MILITARY SPECIFICATION

PRESS, "H" FRAME, VERTICAL, HYDRAULIC, AIR DRIVEN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of floor mounted, air driven, "H" frame, hydraulic press with equipment and accessories specified herein.

1.2 Classification. This specification covers presses of the following sizes. The size press to be furnished shall be as specified (see 6.2.1 and table I).

Size 1 - 25 ton capacity.
Size 3 - 50 ton capacity.
Size 5 - 75 ton capacity.
Size 6A - 100 ton capacity.
Size 7 - 150 ton capacity.
Size 8 - 200 ton capacity.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Design. The press shall be new and of the "H" frame type design capable of performing pressing operations in accordance with the requirements specified herein. The press shall be designed for floor mounting and shall consist basically of a frame, worktable, hydraulic head assembly, hydraulic system, air powered pumping unit, and controls. The press shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the press. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the press shall be graduated in the same system.

3.2.2 Reclaimed materials. The press may contain reclaimed materials provided such materials will not jeopardize the presses intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the press.

3.2.3 Controls. The press shall be equipped with all controls required for complete control of pressing operations. The controls shall provide for rapid advancement of the ram to, and retraction from, the workpiece. Ram speed shall be controllable from zero to maximum rated speed for all operating conditions. All operating controls shall be located convenient to the operator at his normal work station.

3.2.4 Safety and health requirements. All press parts, components, mechanisms, and assemblies furnished on the press, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the press itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

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3.2.5 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The press shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the press at its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the press. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the press shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the press and its related attachments and accessories shall conform to FED-STD-H28.

3.4 Components.

3.4.1 Frame. The press shall be of the "H" frame type fabricated of structural steel sections. The side members shall be rigidly joined together by three or more crossmembers. The upper crossmember shall serve as the headrail to support the hydraulic head assembly. The center crossmember

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shall serve as the worktable. The headrail, and all other crossmembers shall provide rigidity to the frame and maintain parallelism of the side members. The frame shall be supported by a base of adequate length and strength to provide stability during loading at maximum capacity. The frame shall have sufficient accuracy and rigidity to maintain the ram axis perpendicular to the worktable within an accuracy of 0.050 inch per foot.

3.4.1.1 Frame siderails and support pins. The siderails shall be either of the open type or closed type. The siderails shall have a series of equally spaced holes to provide movement of the headrail or worktable. The press siderail holes shall allow either the headrail, or the worktable, or both to be raised or lowered. The hole spacing shall provide a minimum distance between the ram nose and the worktable in accordance with the requirements of table I. The siderails shall support a mechanically operated hoist for raising and lowering the headrail or worktable, as applicable. A set of steel pins shall be provided for supporting the headrail or worktable to the siderails. The pins shall be shaped for easy insertion and removal from the holes. The total shear strength of the pin set shall be not less than double the tonnage rating of the press. The table and table support pins shall be capable of withstanding the bending stresses encountered when operating at 120% of the rated press tonnage without permanent deformation of the table support pins. The pin hole spacing in all members shall be of sufficient accuracy to insure that all pins support equal loads when the pressing force is centered between the headrail and the worktable. The width of the press between the frame and siderails shall conform to the requirements of table I.

3.4.1.2 Worktable. Unless otherwise specified, a standard type worktable shall be furnished. When specified (see 6.2.1), a roll out worktable shall be furnished in lieu of the standard table.

3.4.1.2.1 Standard worktable. The standard worktable shall be designed to directly support the workpiece or the work support devices. The width between the worktable channels on the press shall conform to the requirements of table I.

3.4.1.2.2 Roll out worktable. The roll out worktable, when furnished, shall be provided with tracks mounted to the frame. The roll out worktable shall be easily rolled into and out of its normal work position below the ram axis in order to facilitate loading of the workpiece. The roll out worktable shall be provided with a locking or clamping device for locking the worktable in position below the ram assembly for pressing operations.

3.4.1.2.3 Work support device. The press shall have two work support plates that rest on the table channels to support flat workpieces. The work support plates shall be movable along the entire width of the table to provide the necessary support for any workpiece within the size range of the machine. The upper surface of the plates shall be flat and the bottom surface shall have a tongue, step, shoulder, or other means to hold the plates in alignment to prevent the plates from sliding off the table, but allow

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TABLE I. Sizes and capacities.

Capacities	Sizes					
	1	3	5	6A	7	8
Rated capacity, tons	25	50	75	100	150	200
Width between frame siderails, inches	29	28	35	35	48	48
Width between table channels (max. arbor size), inches	5	7	8	10	12	14
Distance, ram nose to table, inches	36	40	36	37	30	30
Vertical travel of table or headrail, inches	35	30	30	30	20	24
Ram piston travel, inches	6	6	10	9	13	13
Ram travel speed, inches per minute:						
Rapid advance, (at no load)	22	9	6	6	4	3
At full rated load:						
Standard	1-1/4	1-1/2	1	3/4	1/2	1/4
Optional	4-3/4	2-1/4	1-1/2	1-1/4	3/4	1/2

Note: Requirements are not less than those shown.

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free sliding motion on the table channel surface. In addition, a pair of V-block type work support plates, suitable for supporting a round bar during straightening operations, shall be furnished with each press. A single pair of invertable blocks suitable for supporting flat workpieces on one side, and supporting round bars on the other side, may be supplied in lieu of the two pairs of supports required above.

3.4.2 Hydraulic head assembly. The hydraulic head assembly shall contain the hydraulic cylinder, ram piston, ram, and ram extension mechanism. The hydraulic pump may be mounted directly on the head assembly, or may be mounted at any convenient operating position on the press frame. Unless otherwise specified, the head assembly may be either fixed to the center of the headrail or laterally adjustable along the entire operating width of the headrail. When only one type of head assembly is acceptable, the particular type required shall be as specified (see 6.2.1). The movable type head assembly shall be provided with a means for securely clamping or locking the head assembly at any adjusted position within its range of movement. The movable type head assembly shall be capable of performing full rated tonnage pressing operations with the head assembly at any position within its range of travel without deformation of any component or structural member.

3.4.2.1 Cylinder and ram. The cylinder and ram shall be capable of exerting 110% of the full rated press tonnage without damage or permanent deformation. Means shall be provided for rapidly advancing the ram nose to contact the workpiece prior to the pressing operation and for rapidly retracting the ram to the top-of-stroke position after completion of the pressing operation. Unless otherwise specified the ram travel speeds at full load shall conform to the requirements shown as "standard" in table I. When specified (see 6.2.1), the ram travel speed, at full load shall conform to the requirements shown under "optional" in table I. The rapid extension system shall conform to 3.4.2.2 and the retraction system shall conform to 3.4.2.3.

3.4.2.2 Rapid ram advance system. The ram shall have provisions for rapidly advancing the ram to contact the workpiece prior to the pressing operation. Rapid advance may be accomplished by utilizing hydraulic or pneumatic pressure acting directly or indirectly on the ram piston. The average rapid advance speed (total ram stroke length in inches divided by total time in minutes to make a full stroke) shall conform to the requirements in table I. When pneumatic pressure is utilized for rapid ram extension, the hydraulic system shall remain full of fluid during the rapid advance operation, and when normal pressing operations are started, the ram shall immediately start in motion without the necessity for bleeding, or otherwise expelling air from the hydraulic system.

3.4.2.3 Ram retraction system. The ram shall have means for retracting the ram to the top-of-stroke position after completion of a pressing operation and after actuation of the pressure release valve. Ram retraction

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shall be accomplished either by utilizing hydraulic or pneumatic pressure acting directly or indirectly on the bottom of the ram piston, or by utilizing a spring loaded retraction mechanism. Regardless of which system is utilized, pressure release and ram retraction shall be accomplished by operating a single control.

3.4.3 Hydraulic system. The hydraulic system shall consist essentially of a fluid reservoir, a pneumatically powered hydraulic pump, pressure release valve, pressure gauge, hydraulic controls, and necessary piping, hoses, and fittings. The hydraulic system design shall enable the ram to exert any tonnage up to 110% of the maximum rated tonnage specified in table I. All components in the hydraulic system shall have a safe working pressure rating of not less than 120% of the pressure that can be exerted by the pump. The system shall have a pressure relief valve, or similar device, to prevent loading the press beyond 110% of its rated tonnage. Means shall also be provided to prevent extension of the ram beyond its normal "end-of-stroke" position. All components in the system shall be sealed to prevent fluid leakage when the system is operating at full rated tonnage capacity.

3.4.3.1 Fluid reservoir. The fluid reservoir shall be capable of holding not less than 1-1/4 times the volume of fluid required to fill the hydraulic system and force the ram to the bottom of its stroke. If the hydraulic reservoir is pressurized during any part of the operating cycle, such as for rapid ram advance, the reservoir shall have a safe operating pressure rating of not less than twice the pressure that can be imposed on the interior of the reservoir.

3.4.3.2 Hydraulic pump. The hydraulic pump shall be of the pneumatically powered positive displacement type and shall have a pressure and volumetric capacity that will enable the ram to exert 110% rated press tonnage and comply with the ram travel speed requirements of table I. If a reciprocating pump is furnished, the pump shall be either of the double-acting type or of the multi-cylinder, single-acting type. Pressure and displacement fluctuations shall be minimized to the extent that there is no observable variation in ram speed when the pump is operating and a pressing operation is in progress. The pump, or some other part of the hydraulic circuit, shall have a pressure release valve that allows pressure to be released to the reservoir, and allows the ram to be returned to the top-of-stroke position.

3.4.3.3 Pneumatic power system. The power and performance characteristics of the pneumatic motor and other components of the pneumatic system shall be sufficient to operate the press at 110% of maximum rated tonnage capacity as specified herein, when operating on a shop air supply system having a pressure of not less than 80 psi or more than 150 psi. When specified (see 6.2.1), an air pressure regulator, pressure gauge, air filter, water separator, lubricator, and necessary operating valves shall be

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furnished. The pneumatic power system shall be mounted as an integral part of the workhead or as separate units on the press frame.

3.4.3.4 Pressure gauge. The hydraulic system shall include a pressure gauge calibrated to indicate ram force. The gauge shall be legible from the operator's normal work station, and shall not be obstructed by any part of the press.

3.4.3.5 Air bleeding. The hydraulic system shall be either self-bleeding or shall have provisions for bleeding air from the system after the system is filled. After the system is bled, it shall remain free of air under all conditions of operation except draining and refilling of the system.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the press shall be not less than those specified in table I for the press size specified.

3.6 Performance. The press shall be capable of performing routine pressing operations within its rated tonnage capacity. The press shall be capable of exerting a force equal to 110% of its maximum rated tonnage capacity without permanent distortion, or other damage, to any component.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Standard equipment. Unless otherwise specified all standard equipment normally furnished by the manufacturer, shall be furnished. Additional equipment required shall be furnished as specified and fully described (see 6.2.1).

3.8 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.9 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.10 Marking on plates and charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.10.1 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each press. The nameplate shall contain the information listed below. If the press is a special model, the model designation shall include the model of the basic standard press and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

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Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Operating air pressure and volume required
Rated press tonnage.
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.12 Workmanship. Workmanship of the press and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When first article inspection is require, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.5, the test in 4.6.1, and the inspection in 4.7. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

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4.5 Examination. The press shall be visually examined to determine compliance with all requirements of this specification.

4.6 Tests.

4.6.1 Operational test. The press shall be operated and cycled at no load for not less than 10 minutes. Proper operation of all controls, motors, pumps, adjusting mechanisms, and accessories shall be verified during the trial period.

4.6.2 Performance test. Each press shall be loaded and checked for proper operation when loaded to 110% of maximum rated capacity. Each press shall retain the load for a period of not less than one hour. During and after testing, the press shall be examined for permanent distortion, breakage, hydraulic leaks, air leaks, and any other evidence of nonconformance to this specification.

4.7 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Packaging, packing, and marking. Unless otherwise specified, packaging, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. Presses covered by this specification are for use in machine production, repair, and maintenance shops.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2 and table I).

c. First article, if required (see 3.1).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

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f. Additional safety and health requirements, if required (see 3.2.4).

g. Painting, if different (see 3.3.5).

h. Roll out worktable, if required (see 3.4.1.2.).

i. Specify either fixed or adjustable workhead, if required (see 3.4.2).

j. Ram travel speed, if different (see 3.4.2.1).

k. Air pressure regulator, pressure gauge, air filter, water separator, lubricator, and necessary valves, if required (see 3.4.3.3).

l. Size and capacity, if different (see 3.5).

m. Additional performance features, if required (see 3.6.1).

n. Standard equipment, if different (see 3.7).

o. Specify and fully describe optional equipment, if required (see 3.8).

p. Specify repair parts, if required (see 3.9).

q. Nameplate, if different (see 3.10.1).

r. First article inspection, if different (see 4.3).

s. Quality conformance inspection, if different (see 4.4).

t. If packaging, packing, and marking in accordance with MIL-M-18058 is required, specify level required (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operators manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.4 requires compliance only with those OSHA requirements that concern the press itself. It

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does not required compliance with those OSHA requirements that concern the press in its operating environment. OSHA limits the total hazard level of these hazards in the environment and does not limit the hazard level of individual presses in the environment. The procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional press requirements that will integrate the new press into its future operating environment. The press shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial press supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required, because the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1 (f).

6.5 Cross-reference of classification changes. Machine classification changes from previous issues of this specification are as follows:

MIL-P-80149	MIL-P-80149A	MIL-P-80149B
Size 1	Size 1	Size 1
Size 2	Size 2	deleted
Size 3	Size 3	Size 3
Size 4	Size 4	deleted
Size 5	Size 5	Size 5
Size 6	Size 6	deleted
	Size 6A (new size)	Size 6A
Size 7	Size 7	Size 7
Size 8	Size 8	Size 8

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Preparing activity:
DLA-IP

(Project 3442-0040)

Review activities:
Air Force - 84
DLA - GS

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